

Multiple visual relationship forecasting and arrangement in videos [☆]

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ABSTRACT

Visual relationships in videos are often causally connected and time-dependent. The complicated temporal dependency between visual relationships is overlooked in visual relationship modeling. To explore the realistic interactions and model the causality between visual relationships in videos, we propose a new task named Multiple Visual Relation Forecasting and Arrangement (MVRFA). Given a series of frames focusing on a <subject-object> pair, MVRFA aims to forecast the future visual relationships and their arrangement which reflects their temporal interrelationship. To evaluate the MVRFA task, we build a video dataset consisting of 1195 videos with 2666 visual relationships and corresponding arrangements annotated by 5 temporal relational classes. In addition, we present a Multi-task Relationship-centric Transformer model as a baseline, which applies graph convolution to aggregate subject-relationship-object information and uses the relationship multi-query transformer to predict the visual relationships with their arrangement. Experiments on the proposed dataset demonstrate that the proposed baseline achieves superior performance and can better handle the MVRFA task than related models.

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1. Introduction

Images and videos collected from the real world are known to contain quantitative high-level information, such as emotion, relation, and tendency. Therefore, scene and video understanding has attracted many researchers' attention. Many new tasks have been proposed, such as scene graph generation [39,18], captioning [46,6], and visual relation detection [20,43]. The development of visual tasks shows a trend toward specialization. Tasks range from detecting fixed objects [35,7] to understanding connections between more abstract things [45,36], from understanding existing scenarios [33] to more interactive question answering [16,1]. These high-level tasks require modeling the real world and the causality within it. By considering more critical information, they can better interpret the causality.

Causality is the influence of one event on another and the events are called cause and effect, respectively. Some tasks, such as relational reasoning [45], detect the causality between two frames in videos to perform reasoning. They explicitly model and

detect the causality. Other video reasoning tasks require an implicit understanding of causality and predict the future by comprehensively modeling past information. One typical example is visual relation forecasting, which requires modeling past information in video frames and forecasting future visual relationships.

The previous work [22] forecasts the next several visual relationships in sequence and represents one interaction between objects by a single visual relationship. However, due to the varying duration of visual relationships, the relationship between objects can be complicated and diverse. On the one hand, future interactions between objects are often composed of multiple visual relationships which could occur simultaneously. Therefore, one single visual relationship is insufficient to represent the future interaction in a video clip. For instance, in Fig. 1, the baby simultaneously *watches* and *walks towards* the ball. To comprehensively explore future interactions, we should forecast multiple visual relationships in the future instead of the next relationship. On the other hand, the temporal relation between visual relationships is complicated and multiple visual relationships can not be forecasted in sequence. Meanwhile, the combination and order of these visual relationships contain rich information [24]. For example, these visual relationships in Fig. 1 are not in a *grab-lift-watch-towards* sequence, and the temporal relation between *grab* and *hold* shows that the baby is picking up the ball. Therefore, arranging

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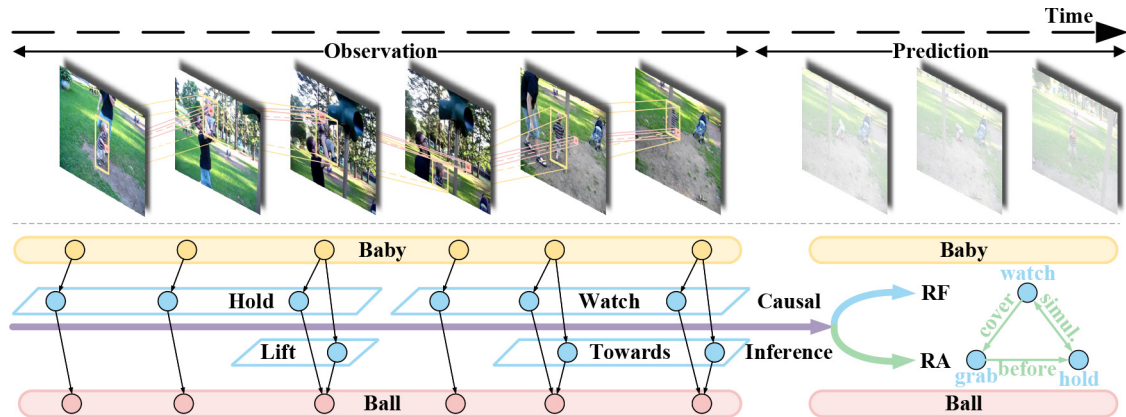


Fig. 1. An example of the proposed task Multiple Visual Relationship Forecasting and Arrangement (MVRFA). The task includes two subtasks: Relationship Forecasting (RF) and Arrangement (RA). Given the observation frames with corresponding visual relationships, MVRFA requires to forecast the visual relationships (RF) and arrange them (RA) with five temporal relational classes: *before*, *cover*, *simultaneous*, *be covered*, and *after*.

these visual relationships to figure out their temporal interrelationship is necessary.

To better model the causality and explore the future interactions between objects, we introduce a task named Multiple Visual Relation Forecasting and Arrangement (MVRFA) in videos. Given a series of frames containing a specific <subject-object> pair, the task is to forecast multiple future visual relationships and arrange them. The future visual relationships can be predicted through the causality connection but the precise temporal locations are infeasible due to the lack of future frame information. Therefore, we utilize representative temporal relational classes to describe the relative arrangement of two visual relationships. Among them, the most common two are *before* and *after*. For example, in Fig. 1, *lift* is *after* *hold*. However, not every arrangement can be described by *before* and *after*. It is obvious that *watch* and *hold* ever occurred at the same time and neither *before* nor *after* are proper to describe their temporal relationship. Under this condition, *simultaneous* is an appropriate description for two events that occurred at the same time. In addition, when *hold* lasts much longer than *watch* so that the duration of *hold* is almost twice of *watch*, *simultaneous* is also not exactly appropriate. Then *cover* and *be covered* are proposed to describe the arrangement. The above five temporal relational classes can fundamentally cover all the possible cases. Although some more specific relational classes can be proposed to further refine the five classes, they would duplicate or overlap with existing relational classes and be too complicated. Considering the comprehensiveness and brevity of the arrangement, we use five temporal relational classes to describe the arrangement: *after*, *before*, *cover*, *be covered*, and *simultaneous*.

As a challenging task, MVRFA requires a full understanding of the observation video clips by exploiting the spatial-temporal information and building the causal links between the past and the future. Additionally, visual relationship forecasting can timely predict future situations in intelligent companion or monitoring, and provide more thoughtful services. Prediction tasks can allow more time for emergency processing than detection tasks.

Since existing datasets do not directly provide videos and annotations for multiple visual relationship forecasting, we reorganize the data and the annotation based on the VidOR dataset [28], providing videos with realistic, swift, and logical visual relationship transitions. We select the 1195 video clips with 34 visual relationship categories on specific <subject-object> pairs from 10000 real-world videos. Compared with other datasets [29,22], the MVRFA dataset focuses on more precise temporal modeling and requires better usage of spatial-temporal information to fore-

cast accurate visual relationships with arrangement and model causality.

To cope with the challenging task, we present a Multi-task Relationship-centric Transformer as a baseline. The model consists of four parts: the spatial modeling module, the temporal encoder, the multi-query decoder, and the multi-task prediction heads. The spatial modeling module fuses visual and semantic features with a graph convolutional network and extracts frame-level features. The temporal encoder encodes the frame-level features into spatial-temporal features with multi-head attention. With a series of learnable relationship queries to extract the spatial-temporal information in the multi-query decoder, two task-specific heads can predict multiple visual relationships and their arrangement simultaneously. Our baseline considers the temporal interrelationship between visual relationships and achieves better performance on both subtasks than related future prediction models through comprehensive experiments.

The main contributions of the paper can be summarized in the following three folds:

- A novel video task named Multiple Visual Relationship Forecasting and Arrangement (MVRFA) is proposed to forecast visual relationships with their arrangement. With the arrangement introducing the temporal dependency between visual relationships, future interactions are described more accurately, facilitating the precise modeling of causality.
- A new dataset based on VidOR [28] is presented for the MVRFA task, which contains 1195 videos annotated with 2666 visual relationships and their arrangement. According to the temporal relative positions of the visual relationships, five temporal relational classes are proposed to specify the concept of the arrangement.
- A Multi-task Relationship-centric Transformer model is presented to simultaneously predict the visual relationships and their arrangement by extracting the global spatial-temporal information with relationship queries, which reflects interactions between visual relationships, and therefore, facilitates multi-task performance.

The rest of this paper is organized as follows. In Section 2, the research background of visual relationship modeling and future prediction are briefly introduced. A specific task definition is given in Section 3. Section 4 describes the formation and statistical information of our MVRFA dataset. In Section 5, our proposed method is presented in detail. Finally, the experiment results are analyzed in Section 6 and the conclusions are given in Section 7.

2. Related Work

2.1. Visual Relation Modeling

The Visual relationship Forecasting task in MVRFA requires modeling the observed video frames and extracting the spatial-temporal information to forecast future interactions. It has great similarity with the visual relation detection task in visual relationship modeling.

The image visual relationship detection models generally consider the global context information [31] besides the local information. To aggregate this information with topology information, graph neural networks are used to model frame-level relational triples [26]. In addition to considering the relationship in triples, researchers introduce the interrelationship between visual relationships [25] or establish relationships between triples through a hierarchical graph structure [21] to improve detection accuracy.

Compared with the visual relationship detection task on images, the task on videos is more challenging since the videos contain more information than images. VidVRD [29], Action Genome [13], and VidOR [28] present dense annotations of visual relationships and object trajectories. To capture the dynamic data in videos, some researchers [27] apply 3d convolution in video visual relationship detection. However, 3d convolution requires a lot of computation and there is much redundant information in the video. To solve the problem of sparse information, TSPN [37] uses a new search method, which improves efficiency while maintaining its effectiveness. Cao et al. [3] propose to use 3d proposals to combine three different sub-tasks and share the feature representation for efficiency. On the other hand, Li et al. [17] found that VidVRD datasets also suffer from long-tail distribution and aimed to improve the robustness of the model.

Temporal information also matters in video visual relation modeling. To build the bridge between the temporal information and the semantic information, Xiao et al. [38] propose a novel task called vRGV and perform spatial temporal localization in the video with given relations. Our MVRFA, which is opposite to vRGV, further utilizes the temporal links to better predict visual relationships.

2.2. Future Prediction

As a subtask of future prediction, visual relationship forecasting, like most future prediction tasks, focuses on the causal connection between the past and the future.

Early future prediction tasks, such as series prediction with Markov models [9] and filtering models, do not focus on causality but on the nature of the data due to the limited computing power. Later, more complicated tasks, such as multivariate stochastic process prediction [2], traffic flow prediction [12], and action prediction [23,19], developed. It indicates that the current state-of-the-art model can extract features from various data types and start to focus on causality.

As causality modeling gets more attention, more and more inference tasks and datasets are proposed and have different representations. VCR [41] constructs the reasoning task with multiple-choice questions and answers, which requires combining text and image information and figuring out object relationships in images. Similarly, our MVRFA task focuses on object relationships in videos and considers more temporal casual connections, such as past-future relationships and visual relationship arrangement. TRN [45] is designed to learn temporal dependencies between video frames and perform activity recognition tasks while MVRFA explores the temporal dependencies between visual relationships

and predicts future interaction between pairwise objects. Both these two tasks depend on temporal relational reasoning, which is the basis of future prediction, except that TRN recognizes coarse-grained activities while our MVRFA requires more challenging fine-grained temporal information.

3. Task Definition

The new task, multiple visual relationship forecasting and arrangement (MVRFA), aims to perform causality reasoning and interaction forecasting based on spatial-temporal information. The task has two subtasks: relationship forecasting (**RF**) and relationship arrangement (**RA**). The goal of **RF** is to forecast multiple visual relationships in the future. On this basis, **RA** intends to arrange these visual relationships. The specific definitions are as follows:

Relationship Forecasting: given a series of frames with a specific <subject-object> pair and the corresponding visual relationships, **RF** aims to forecast visual relationships $r = \{r_1, r_2, \dots, r_N\}$ in the future by modeling the causality connections between the future and the past visual relationships. As shown in Fig. 1, given a series of observation images and the historic interactions, the adult *watches*, *holds*, and *lifts* the baby, **RF** should forecast the future visual relationships between the adult and the baby, *watch*, *pat*, and *hug*.

Relationship Arrangement: assuming we know the future visual relationships $r = \{r_1, r_2, \dots, r_N\} \in R$, this subtask aims to arrange these visual relationships. Defined 5 temporal relational classes, before, cover, simultaneous, be covered, and after, **RA** aims to predict the temporal relation s_{ij} between each visual relationship pair r_i and r_j .

4. MVRFA Dataset

4.1. Data Collection

4.1.1. Video Selection

Our MVRFA dataset is generated based on the VidOR dataset. The VidOR dataset contains 50 relationships. Among these relationships, 42 relationships, containing rich dynamic information, are called contact relationships. The other relationships describing relative spatial positions between object pairs and representing their co-occurrence are called spatial relationships. The videos with continuous contact relationships are considered to have dynamic interactions and strong causality links. To better train and evaluate our task, we need to filter out those video clips from the VidOR dataset.

More specifically, we list all the <subject-object> pairs and their corresponding visual relationships in the videos. Then the videos will be split into two parts if they contain more than 1400 frames or if there are more than continuing 150 frames without spatial relationships in the videos, as too long videos have weak casualty links and we hope the object pairs co-occur in the video to provide enough visual information. In addition, videos with less than three contact relationships will be deprecated. Under these conditions, we selected 1325 qualified video clips with 34 visual contact relationships and 59 object categories.

4.1.2. Video Split

Before annotating the videos, we have to find a boundary to split every qualified video into two parts: observation clips and prediction clips. In this process, we follow three fundamental principles:

- The boundary should be put between two dispersed visual relationships and one visual relationship should preferably not be divided into two parts.
- The length of the prediction clips should be longer than the duration of a single visual relationship but no longer than the length of the observation clips.
- The prediction clips should not end with some visual relationships representing static interactions, such as *watch*, or they will be cropped.

After dividing these videos into the observation part and the prediction part, we count the duration of the observation clips. The durations shown in Fig. 2a vary widely, and the density of the visual relationships in different video clips consequently changes. To evenly take keyframes of every visual relationship, we apply two schemes to take 16 or 32 keyframes in a video. Additionally, we only select the frames where the subject and the object coexist.

For every keyframe, we present the bounding box of the subject and the object to assist in modeling the interaction. In addition to the contact relationships that represent dynamic interactions, the spatial relationships representing the relative position of two objects are also given to help understand the scene.

4.1.3. Annotation

For the **RF** subtask, we present the label of the visual relationships that appear in the prediction video clips. To represent the temporal arrangement of two visual relationships r_i and r_j in the **RA** subtask, we propose five temporal relational classes: *before*, *cover*, *simultaneous*, *be covered*, and *after*. The specific rules of division between the five classes are introduced below.

Let us denote the duration of r_i and r_j by t_i and t_j , respectively. We first calculate the duration t_{ij} of the overlapping part of r_i and r_j by $t_{ij} = t_i \cap t_j$. Then we divide t_{ij} by t_i and t_j , respectively, and get the ratio α_i and α_j by $\alpha_i = t_{ij}/t_i$ and $\alpha_j = t_{ij}/t_j$.

Subsequently, we can represent the temporal relationship between two visual relationships with α_i and α_j . Since the temporal relationship should be divided into three classes (*before/after*, *cover/be covered*, *simultaneous*), we use (α_i, α_j) to represent the coordinate of a point on a 2-dimension plane and assign each point into one of three clusters. To select the optimal thresholds, we adopt the same strategy as the K-means algorithm and try to minimize the intra-class distances of the clusters. Furthermore, we

compare the sum of the intra-class distances under different threshold conditions within the range of 0–1 and a step of 0.05, finding that 0.5 and 0.55 are the optimal thresholds.

More specifically, if both α_i and α_j are greater than 0.55, the temporal relational class is *simultaneous*. If both α_i and α_j are less than 0.5, the relationship is *before* or *after*. In other cases, it indicates that one visual relationship *covers* another one, and another visual relationship is *covered* by this visual relationship.

4.2. Dataset Division

To reasonably divide these video clips into the training and the testing set, we count the frequency of the occurrence of the visual relationships in the prediction video clips. The statistical results are shown in Fig. 2b, which shows a long tail distribution. To enable the testing set to test the ability of the model comprehensively, we hope the training set retains all visual relationship categories while mitigating the long-tail effect in the testing set so that the testing set has a more balanced distribution.

Based on the above requirements, we first discard part of the video clips that only contain high-frequency visual relationships to alleviate the long tail distribution in the dataset. Then we take about 8 to 16 percent of videos containing specific visual relationship categories. For the videos containing visual relationship categories with lower frequencies, we take a higher proportion (8%) to put into the testing set and vice versa. The final dataset contains 1195 videos with 1075 videos in the training set and 120 video clips in the testing set. The comparison of the distribution of visual relationship categories between the training set and the testing set is shown in Fig. 3.

4.3. Statistics

4.3.1. Video Statistics

The proposed dataset consists of 1195 video clips with 59 object categories and 34 visual relationships. The video clips last at most 1382 frames and at least 101 frames. These 59 object categories are combined into 125 kinds of <subject-object> pairs. In the 1195 video clips, there are 2666 visual relationships in the unknown future to be predicted, resulting in an average of 2.23 visual relationships in a video. And the spread of the visual relationships is shown in Fig. 2b.

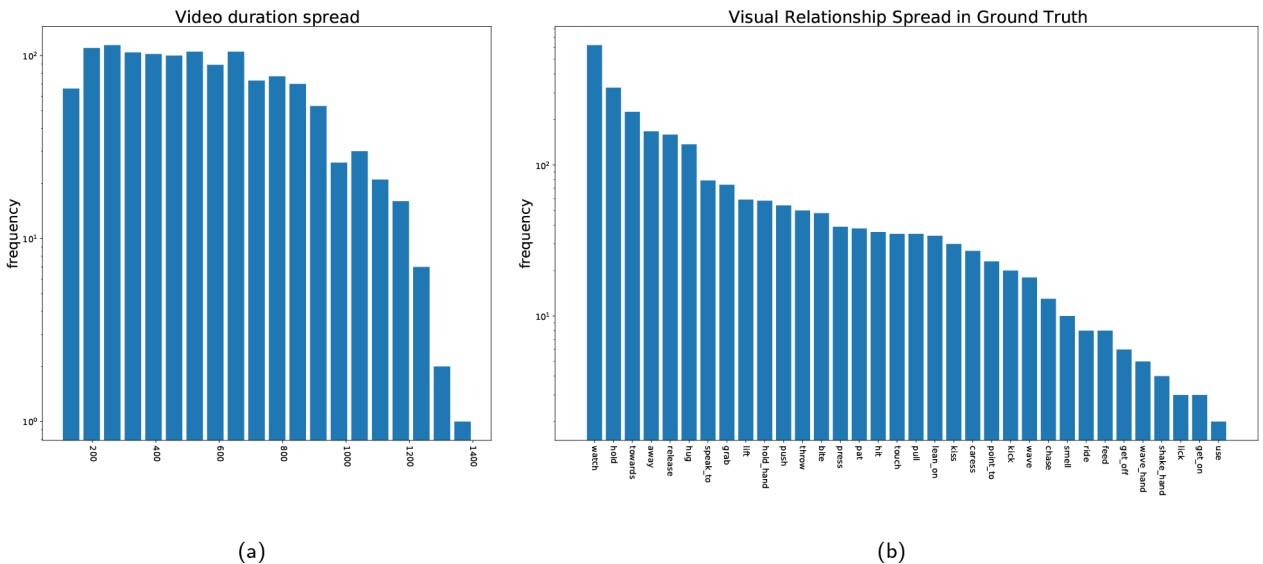


Fig. 2. (a) The duration spread of the observation video clips. It varies in a wide range from 100 frames to more than 1300 frames. (b) The frequency spread of all visual relationships in the ground truth of our dataset.

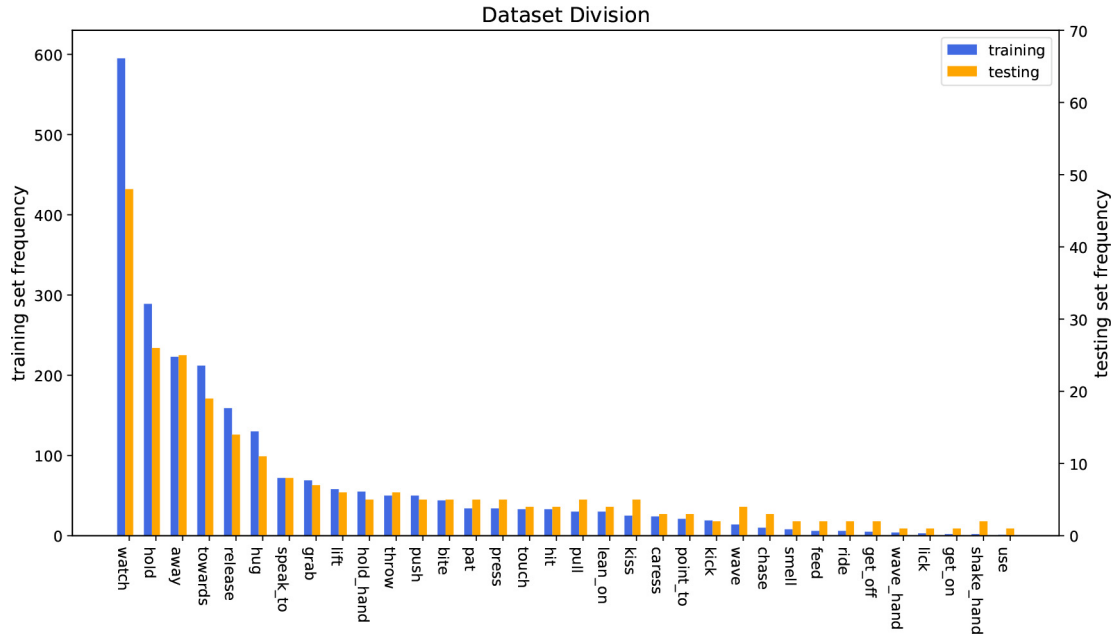


Fig. 3. The frequency of the visual relationships in the training set and the testing set.

4.3.2. Arrangement Statistics

Our MVRFA task is based on causality which affects the transition and temporal interrelationship between visual relationships. To initially verify the existence of this causality, we explore the temporal dependencies between the visual relationship in the prediction video clips.

Firstly, we select 10 visual relationships with a moderate frequency to facilitate visualization. Then we count the pairwise occurrence among them. Since we pay attention to the temporal dependency between pairwise visual relationships, we further count the distribution of the temporal relational classes for each visual relationship pair. With the distribution, we can judge the most frequent temporal relational class between specific visual relationship pairs. To figure out whether there is a specific temporal dependency, we calculate the variance of each distribution. In addition, for the pairwise visual relationships that never appear together, the variance is marked as 0 which symbolizes there are no temporal dependencies between them. The final variance map of these distributions is shown on the left side of Fig. 4. In the figure, the variance of many distributions reaches a maximum of 0.16, showing an obvious statistical pattern. For example, the variance of the distribution between *hold_hand* and *release* reaches 0.16, reflecting there is a strong temporal dependency between them.

On the other hand, we present the most frequent temporal relational classes in each distribution on the right side of Fig. 4. For example, *release* tends to occur after *lift* or *grab*, which is consistent with our common experience. In addition, some visual relationships are not in the order of *before* and *after*, but their arrangement still has a strong statistical law. For example, *pat* often coexists with other three words: *bite*, *lift*, and *hold_hand*. The results indicate that the other three temporal relational classes, *cover*, *be covered*, and *simultaneous*, are also widespread.

5. Multi-task Relationship-centric Transformer

5.1. Framework

Our proposed baseline for the MVRFA task can be divided into four parts. The overall structure of the proposed method is shown

in Fig. 5. The spatial modeling module fuses different features of the subject, the object, and the visual relationships into nodes in a relation graph to model spatial information. The temporal encoder applies Multi-head Attention to establish temporal connections between frame features. The multi-query decoder uses relationship queries to extract visual relationship features from spatial-temporal information. The multi-task prediction heads predict future visual relationships and their arrangement and employ the optimal assignment algorithm to establish the correspondence between the predictions and the ground truth.

5.2. Data Preprocessing

We process the data into three kinds of features: the visual feature, the semantic feature, and the spatial feature. Specifically, we crop out the subject and object parts of the image according to their respective bounding box. To represent the visual relationships, we crop the part of the image surrounded by the minimum bounding box of the subject and the object. These images are then fed into a pre-trained Inception-ResNet [32] to extract 1536-dimensional features as the corresponding visual features. As for the semantic feature, we apply a pre-trained Word2Vec model to extract a 300-dimensional feature for every visual relationship and object category. For spatial features, we follow the previous work [11] to extract a 20-dimensional feature to describe the spatial relationship between the subject and the object using their bounding box.

5.3. Spatial Modeling

Spatial modeling aims to construct frame-level features with graph convolution for every keyframe. In each frame, the subject, the object, and every visual relationship have their corresponding visual features f_v and semantic features f_s . The spatial feature f_{sp} is generated at the frame level. Since semantic features contain much complementary information to visual features, we use semantic features to assist in extracting high-level information from visual features. We map visual and semantic features to the same dimension with two linear layers and concatenate them to

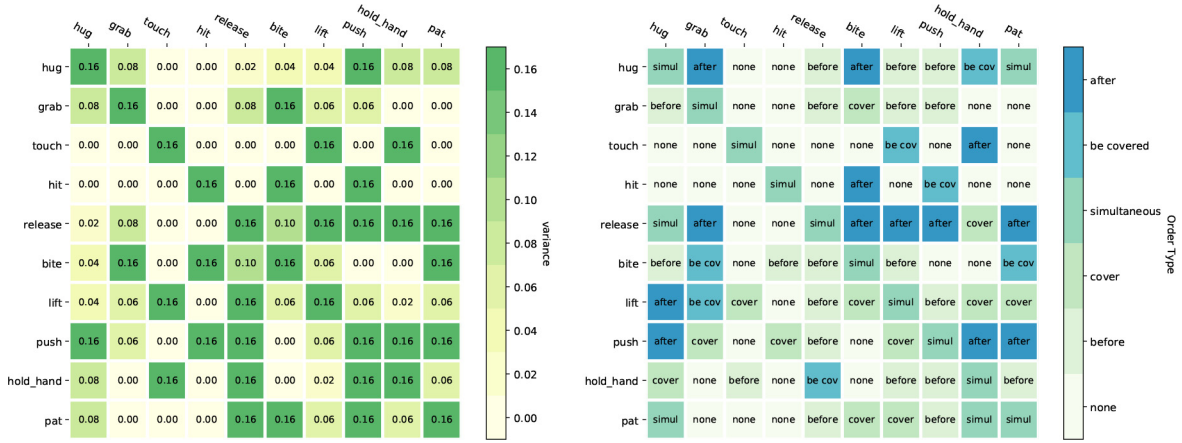


Fig. 4. Variance distribution of different temporal order (left) and the most frequent temporal relational class (right) between visual relationships. The temporal relational class between visual relationships with a variance of zero is denoted as none. The results show the dependency between visual relationships and represent obvious statistical regularity.

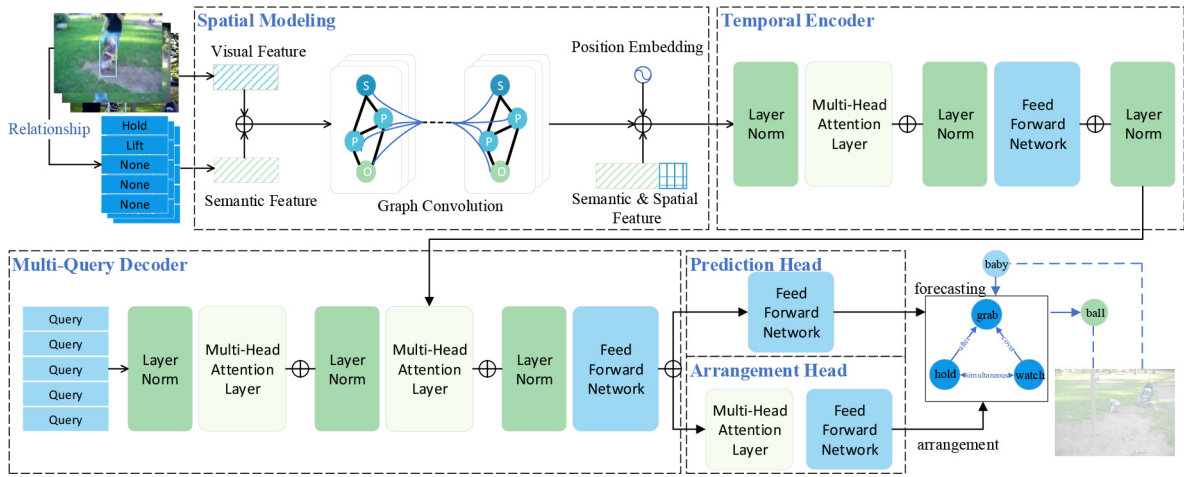


Fig. 5. The overall structure of our proposed baseline. The model is composed of four parts: the spatial modeling module, the temporal encoder, the multi-query decoder, and the multi-task prediction heads.

represent node features, $X = \text{Concat}([f_v, f_s])$. Since there might be multiple visual relationships in a frame, we build nodes for each visual relationship. After obtaining the nodes for the subject, object, and visual relationships, we perform graph convolution on them which can be represented as:

$$\tilde{X} = \sigma(\tilde{L}_{\text{sym}} X W) = \sigma(D^{1/2} \tilde{A} D^{-1/2} X W) \quad (1)$$

where $\tilde{X}$ is the node features after graph convolution. σ represents the Sigmoid activation function. $\tilde{L}_{\text{sym}}$ is the renormalized Laplacian and W is a learnable parameter. D denotes the degree matrix of the graph. $\tilde{A}$ is an autocorrelation adjacency matrix equal to $A + I$ where I is the identity matrix and A is the adjacency matrix that contains the topology information.

In the adjacency matrix of a single frame, the subject and object nodes are connected through the visual relationship nodes. To utilize the interrelationship between visual relationships, we also create connections between visual relationship nodes. While constructing an adjacency matrix of a video, we follow the same principle as in a single frame, prohibiting the connection between the subject nodes and the object nodes, but allowing them to transmit their state information across time. The adjacency matrix of videos is independent of time and only relates to the type of nodes.

In a word, two nodes are not connected only if they represent the subject and the object respectively.

To distinguish the nodes of objects and predicates, let them have the following representations: $X = \{X_i^\alpha\}_{i=1}^{\alpha=S,O,P}$. Among them, S, O , and P , the types of the node, indicate the subject, the object, and the predicate, respectively. The connectivity A_{ij} between X_i^α and X_j^β can be represented as:

$$A_{ij}(X_i^\alpha, X_j^\beta) = \begin{cases} 0, & \alpha = S, \beta = O \\ 1, & \text{others} \end{cases} \quad (2)$$

where 0 indicates X_i^α and X_j^β are not connected and 1 is the opposite.

Since $\tilde{X}, f_s$, and f_{sp} all contain frame-level information, we concatenate them together to construct the frame-level features F_T :

$$F_T = \text{Concat}[\tilde{X}^S, f_s^S, \tilde{X}^O, f_s^O, \tilde{X}^P, f_s^P, f_{sp}] \quad (3)$$

where $\tilde{X}^S, \tilde{X}^O, \tilde{X}^P$ and f_s^S, f_s^O, f_s^P represent the node features and semantic features corresponding to the subject, the object, and the predicate, respectively.

5.4. Temporal Encoder

We apply a transformer [34] encoder structure with the multi-head self-attention mechanism to establish temporal connections between frame-level information. With the temporal encoder, we can obtain the spatial-temporal features, X_{st} . The attention mechanism can be represented as:

$$Attention(Q, K, V) = \sigma \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

where Q, K, V above represent query, key, and value, respectively. σ represents the Sigmoid activation function. In the temporal encoder, Q, K, V are the same features extracted from the frame-level features F_T .

5.5. Multi-Query Decoder

To implement the multiple visual relationship forecasting tasks, we apply the multi-query decoder with reference to DETR [4]. To forecast multiple independent visual relationships, we set up a fixed number of learnable queries: $Q = \{Q_i\}_i^N$. Considering that the decoder is permutation-invariant, learnable embedding parameters are added as positional weights to differentiate these queries. Then we used the relationship queries to extract N features representing visual relationships $X_{vr} = \{X_{vr}^i\}_i^N$ from the spatial-temporal features X_{st} with the multi-head attention:

$$X_{vr} = Multi-headAttention(Q, X_{st}, X_{st}) \quad (5)$$

5.6. Multi-task Prediction Heads

After obtaining the relationship features X_{vr} , we use two branches to predict visual relationships and arrange them, respectively. For relationship forecasting, we use an MLP as a **RF** head to map visual relationship features to label probabilities.

As for the **RA** task, the arrangement of visual relationships is presented as a cross-correlation matrix $\hat{\mathcal{S}}^{N \times N}$ between the N visual relationships. To model the temporal dependency between visual relationships, we refer to the multi-head attention structure. With the linear layers, we obtain the query, key, and value of the N heads which represent the temporal association between one visual relationship and all N visual relationships as below:

$$\begin{cases} Q_i = X_{vr}^i W_i^Q \\ K_i = X_{vr}^i W_i^K \\ V_i = X_{vr}^i W_i^V \end{cases} \quad (6)$$

where W_i^Q, W_i^K, W_i^V are the parameters of the linear layers to process X_{vr}^i , the feature of the i_{th} visual relationship, into the query, key, and value of the attention mechanism. Then we apply the attention mechanism to construct temporal association between different visual relationships as Eq. 7.

$$head_{ij} = Attention(Q_i, K_j, V_j) \quad (7)$$

where $head_{ij}$ can represent the relative temporal arrangement of the i_{th} visual relationship and the j_{th} visual relationships. To obtain the cross-correlation arrangement probability matrix, we concatenate the attention head and map the features into arrangement probabilities corresponding to five temporal relational classes with a learnable matrix W_A as below:

$$\hat{\mathcal{S}}^{N \times N} = Concat \begin{pmatrix} head_{11} & \dots & head_{1N} \\ \dots & \dots & \dots \\ head_{N1} & \dots & head_{NN} \end{pmatrix} W_A \quad (8)$$

Since the forecasting result of a video contains the probability of every predefined visual relationship, we need to find the optimal correspondence between each prediction and the ground truth before calculating the loss function.

The model forecasts N visual relationships and N is often more than the number of visual relationships in the ground truth, n . To establish a one-to-one correspondence, we add the none type into the ground truth to increase the number of the visual relationships from n to N referring to DETR [4].

Let us mark $\hat{y} = \{\hat{y}_i\}_{i=1}^N$ as the predicted labels and y as the ground truth. Then we use y_{ij} to denote the probability that the i_{th} prediction $\hat{y}_i \in \hat{y}$ corresponds to the j_{th} ground truth $y_j \in y$. The optimal allocation is to make the corresponding probability y_{ij} of every prediction $\hat{y}_i$ as large as possible. If we denote by $\mathcal{L}_{cost}(\hat{y}_i, y_j) = 1 - y_{ij}$ the match cost, this problem can be equivalent to finding an allocation $\hat{\sigma} \in \zeta_N$ with the lowest cost:

$$\hat{\sigma} = \arg \min_{\hat{\sigma} \in \zeta_N} \sum_i^N \mathcal{L}_{cost}(\hat{y}_i, y_{\sigma(i)}) \quad (9)$$

where ζ_N refers to all the mapping sets between the predictions and the ground truth. Referring to the previous work [30], we use the Hungarian algorithm to quickly get the optimal assignment between the predictions and the ground truth.

For both the **RF** task and **RA** task, we rearrange the predictions $\hat{P}^N$ and $\hat{S}^{N \times N}$ with the correspondence $\hat{\sigma}$ and then compute the loss function with the cross-entropy loss in Eq. 10:

$$\begin{cases} \mathcal{L}_{RF} = CE(\hat{P}_{\hat{\sigma}}^N, P^N) \\ \mathcal{L}_{RA} = CE(\hat{S}_{\hat{\sigma}}^{n \times n}, S^{n \times n}) \end{cases} \quad (10)$$

where P^N and $S^{n \times n}$ represent the ground truth of the visual relationship and the arrangement matrix, respectively.

6. Experiments

In this section, we first introduce the experiment details in Section 6.1. In Section 6.2, we conducted experiments on the proposed MVRFA dataset to compare the performances of our method with other related methods. In Section 6.3, ablation experiments are conducted to validate the effectiveness of the extracted features and the multi-query structure. We discuss the compatibility of the two tasks and perform qualitative analysis in Section 6.4 and Section 6.5, respectively.

6.1. Implemental Details

We train the two subtasks, **RF** and **RA**, with two Adam [14] optimizers, separately. Their learning rates are respectively set to $2e-4$ and $8e-6$. To balance the training process of these two subtasks, we optimize their parameters simultaneously with the total loss $\mathcal{L}$ equal to the weighted sum of $\mathcal{L}_{RF}$ and $\mathcal{L}_{RA}$. Since $\mathcal{L}_{RF}$ is almost twice as large as $\mathcal{L}_{RA}$, we set the weight of **RF** and **RA** to 1 and 2, respectively. The total loss can be calculated as follows:

$$\mathcal{L} = \mathcal{W}_{RF} \mathcal{L}_{RF} + \mathcal{W}_{RA} \mathcal{L}_{RA} \quad (11)$$

where $\mathcal{W}_{RF}$ and $\mathcal{W}_{RA}$ are the weight of the **RF** loss and the **RA** loss.

In the training process, we adopt the early stopping strategy. More specifically, we end the iteration while the total loss of the testing set does not decrease for 30 consecutive times.

For evaluation metrics, we use accuracy and AP as indicators for visual relationship forecasting. Since the model generates multiple visual relationship predictions and the predicted probabilities from different predictions are mutually exclusive, we directly

select the visual relationship label with the largest probability from every prediction, and the probability is used to represent the corresponding confidence. Then we can compute the accuracy and AP using the predicted labels and confidences. Similarly, the accuracy of the off-diagonal elements with the largest probability in the arrangement matrix is the indicator for arrangement prediction.

6.2. Quantitative comparison with Other Methods

Since there is no ready-made method to handle the MVRFA task, we select and modify several related methods for comparison.

First, we introduce some classical autoregressive temporal prediction models, which are widely used in future prediction fields.

- **LSTM** introduces gated units based on traditional RNN to preserve long-term and short-term memory and obtain better performance in longer sequences [10].
- **GRU** refines output by controlling the flow of information through the model with an update gate and a reset gate. Compared with LSTM, GRU has fewer parameters and similar performance [5].
- **Transformer** utilizes the self-attention mechanism which provides context for any position in the input sequence. Unlike RNNs, it processes the entire input simultaneously and is more efficient [34].

It is worth attention that the autoregressive model predicts the following N visual relationships with intrinsic order (*before* and *after*). We do not want to add the arrangement head to destroy their original structures. The effect of adding an arrangement head after the autoregressive model is shown in Section 6.3.2.

Apart from the above autoregressive models, we also provide some well-designed future prediction models which apply graph convolution operation and match our task well.

- **ASTGCN** combines the spatial-temporal attention mechanism, graph convolution, and standard convolution to describe the temporal futures [8].
- **TGCN** introduces the GCN and GRU to learn complex topological structures and capture the spatial-temporal dependence [44].
- **STGCN** formulates the traffic flow task on graphs and builds the model with complete convolutional structures [40].

The above models are all state-of-the-art graph-based traffic flow prediction methods, which predict the future states of these nodes in the graph. We use the nodes to represent visual relationship features and feed them into the prediction and arrangement heads.

Similar to other visual relationship datasets[15], as shown in Fig. 2b, our dataset suffers from serious long-tail distribution. To evaluate the models' performance more comprehensively, we also show the performance of a random baseline and a frequency baseline built off training set statistics with reference to Neural Motifs [42]. The random baseline generates random probabilities for each visual relationship and temporal relational class. The frequency baseline, **FREQ**, looks up the distribution of relationships between the known subject and object types computed in the training set and selects the most frequent visual relationships. For the unseen subject-object pair, the model directly outputs the most frequent visual relationships among all object pairs. Since the frequency-based model does not predict probabilities of visual relationships, we do not calculate AP on this model. With regard to the arrangement task, we utilize the most frequent temporal relational class between the two predicted visual relationships as the final arrangement prediction.

Table 1 shows the effectiveness of our method in the proposed task. These classical temporal modeling methods share our feature extraction and output modules but have obvious performance differences with our model. Since visual relationships are temporally continuous and all last for a period of time. However, the classical temporal modeling methods in an autoregressive manner ignore the durations and temporal interrelationships of visual relationships and therefore simplify the causality connection which is the basis of the MVRFA task.

For traffic flow prediction models, most of the methods perform poorly. This is because traffic flow prediction models focus on mid-and-long term cyclical information while visual relationships in videos require more attention to short-term causality connections. Moreover, in MVRFA, the status of some nodes in the graph has no information since they represent no visual relationships and there would be meaningless information flow between the same nodes in different time steps. However, the traffic flow methods focus on continuous node status transition which leads to their poor performances. Our method and other autoregressive methods without graph structure concentrate on the semantic meaning of the whole graph to model the interaction between <subject-object> pair. However, the idea of node-level embedding is worth learning in visual relationship modeling.

6.3. Ablation Study

6.3.1. Effectiveness of multiple features

The proposed method models visual relationships with visual, semantic, and spatial features. To verify the effectiveness of these features, we conducted ablation experiments with four experimental settings: visual, semantic, visual + semantic, and full. The four experiments only utilize corresponding features to construct frame-level features.

The results in Table 2 show that all of these three features contribute to our proposed task. Visual information can reflect implicit detailed information, such as the state of the object, which can provide action tendencies for future prediction. And semantic features have explicit information about visual relationships and help to understand actions. It is obvious that the prediction performance

Table 1

Comparison with related methods. The result shows the effectiveness of our method in the MVRFA task.

Model	Forecasting		Arrangement
	Acc	AP	Acc
RANDOM	11.67	8.92	25.00
TGCN [44]	31.27	24.32	26.74
STGCN [40]	39.17	25.17	33.43
ASTGCN [8]	41.46	28.30	25.10
GRU [5]	43.84	28.36	22.02
LSTM [10]	45.83	30.04	23.08
Transformer [34]	55.81	37.06	23.06
FREQ [42]	55.63	-	30.23
Ours	58.43	43.17	40.31

Table 2

Ablation study on input features. The full setting combines visual, semantic, and spatial features.

Feature	v	se	sp	Forecasting		Arrangement
				Acc	AP	Acc
visual	✓			44.38	30.28	33.53
semantic	✓			50.62	31.39	35.08
vs	✓	✓		53.94	36.68	31.98
full	✓	✓	✓	58.43	43.17	40.31

Table 3

Comparison between multi-query and autoregressive structure. 'Autoregressive' represents the model with autoregressive structure and multi-task prediction heads.

Structure	Forecasting		Arrangement
	Acc	AP	Acc
autoregressive	53.30	37.25	19.96
autoregressive*	50.89	33.35	30.52
multi-query	58.43	43.17	40.31

Table 4

Comparison between different training settings. **RF** represents only training the visual relationship forecasting module, and **RA** represents only training the visual relationship arrangement module. **RF + RA** symbolizes training the whole model simultaneously.

task	Forecasting		Arrangement
	Acc	AP	Acc
RF	53.24	36.50	-
RA	-	-	29.94
RF + RA	58.43	43.17	40.31

of the visual + semantic setting outperforms the settings with only visual or semantic features.

Spatial relationships help to understand the historical positional relationship between subjects and objects. The full setting achieves better results in both tasks compared with the visual + semantic setting.

6.3.2. Effectiveness of multi-query structure

To simultaneously predict future visual relationships and their arrangement, we propose to use the multi-query structure in this paper. The previous work [22] performs visual relationship forecasting in an auto-regressive manner. In order to make a fair comparison with our method, we only replace the multi-query structure in the model with an autoregressive structure and keep other structures unchanged. The autoregressive structure constructs a query for each observed keyframe and predicted a time step. For the observed keyframes, the model represents the query by the ground truth of the prediction labels. For the predicted time step, the model constructs the query with a zero vector. Then the model encodes the queries with positional embedding and outputs predictions for all the time steps with cross attention. The autore-

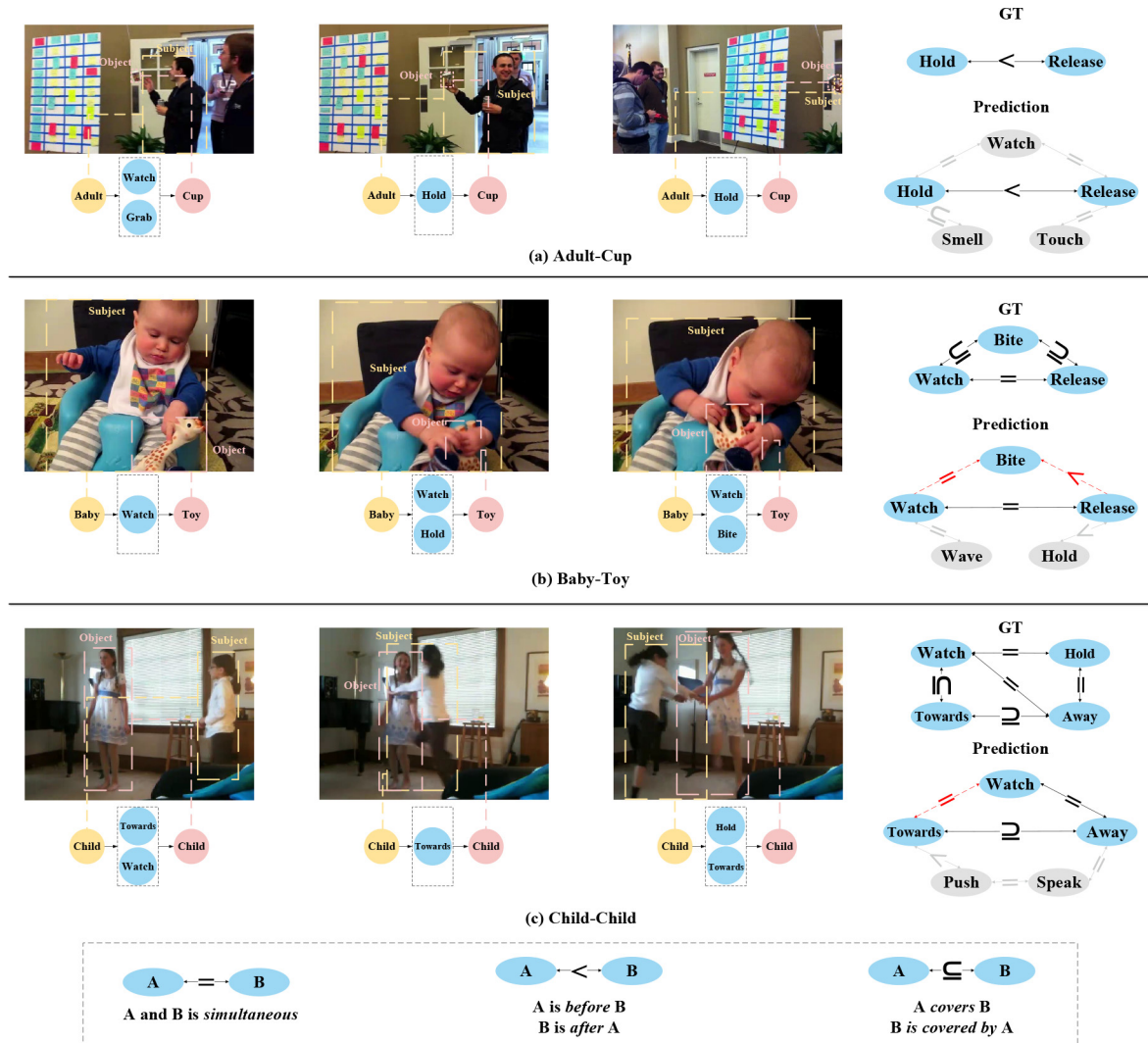


Fig. 6. Visualization examples of the input images, the ground truth, and the prediction results. The yellow bounding boxes mark the subjects and the red bounding boxes mark the objects in the input images on the left side of the figure. On the right side of the figure, the blue ellipses represent the correct visual relationships and the grays represent these not occurring in the ground truth.

gressive method learns from all the previous observations but arranges every prediction in a fixed order. The model is marked as 'autoregressive' in Table 3.

The predicted visual relationships of the autoregressive structure have an intrinsic order, which means the visual relationship of the later time step occurs after the visual relationship of the previous time step. However, as mentioned earlier, 'before' and 'after' are improper to describe the temporal dependency between some visual relationships. To exclude the interference of the default order, we add our arrangement head after the autoregressive model and evaluate its performance. This model is marked as 'autoregressive*' in Table 3.

The results in Table 3 show that our multi-query structure has significant improvements over the autoregressive one. On the one hand, the default order of the autoregressive method is far from the real arrangement. On the other hand, the label prediction of the autoregressive model is also affected. The autoregressive model contains a precondition that the forecasting results have a clear order like *hold-hug-release*. However, visual relationships all last for a period of time, and their temporal dependency on each other is more complicated than this precondition. Therefore, it requires predicting multiple visual relationships and their arrangement without presupposing order. Our model with the multi-query structure generates multiple predictions simultaneously. The arrangement is not presupposed but implied in the features and can be extracted with the arrangement head. As a result, our model not only performs well in the arrangement task but also achieves improvement in the Forecasting task.

6.4. Compatibility of multiple tasks

Intuitively, we have to judge the categories of multiple visual relationships before arranging them. And there would be a simple question: Are these two tasks compatible? We intend to design some experiments to validate this compatibility in this section.

In the previous layers of the model, these two tasks shared features. The forecasting head and the arrangement head differentiate the shared features and output the predictions. If the two tasks are not compatible, the training target of these two subtasks would be contradictory and the performance of the multi-task learning strategy would be worse than the single-task one. On the contrary, two compatible tasks may require common low-level features and promote the extraction of shared features. Then the results of learning the two tasks simultaneously will be better.

In the first two experiments, we train one output head along with the feature extraction layers and freeze the other one. The **RF** setting trains the forecasting head and freezes the arrangement head and the **RA** setting is the opposite. In the third experiment called **RF + RA**, the two tasks share the feature extraction layers and are trained simultaneously.

Table 4 shows the result of different training settings. The performance of **RF + RA** setting significantly outperforms another two settings. The improvement indicates strong compatibility between the two tasks.

6.5. Qualitative analysis

Fig. 6 demonstrates some visualization examples of the input images and the prediction results along with the corresponding ground truth. In the input images, we mark the subjects with yellow bounding boxes and the objects with red bounding boxes. On the right side of the figure, we denote all the visual relationships that occur in the ground truth by the blue ellipses and the other visual relationships by the gray ellipses. As for the arrangement, we utilize three symbols shown at the bottom of the figure to represent these temporal relational classes. Furthermore, these symbols that are

correctly predicted are in black, the wrong arrangement predictions are in red, and the unknown predictions are in gray.

From the results, we can see that our model can forecast most of the visual relationships and give a rough estimation of relative temporal arrangement. However, some prediction results will be biased towards triplets with high frequency, such as *<child-speak_to-child>*, reflecting that our model may be affected by long-tailed distribution data.

7. Conclusion

This paper introduces a new task named Multiple Visual Relationship Forecasting and Arrangement (MVRFA) in videos, which is defined to predict visual relationships and their arrangement with five specific temporal relations given a series of frames with a specific *<subject-object>* pair. Correspondingly, we introduce a new dataset annotated with 2252 visual relationships and their arrangement in 1195 videos. In addition, we propose a baseline method called Multi-task Relationship-centric Transformer, which simultaneously forecasts multiple visual relationships and arranges them. It aggregates information in semantic and visual features using a graph structure, extracts global information with relationship queries, and predicts multiple visual relations and arrangements with the forecasting head and the arrangement head simultaneously. Experiments are conducted to show the effectiveness of our baseline.

Above all, the MVRFA task links the previous observations and the current relationships by causality in videos. The predicted relationships can be important cues in VidVRD. However, most methods in VidVRD focus on the current frame and ignore the relationship detection results in the previous frames. It should be considered to combine all the historical detection results to assist in predicting the current visual relationships. Furthermore, it is also necessary to consider the temporal arrangement between visual relationships, since the VidVRD task requires more precise temporal relationship localization.

Although our method achieves considerable performance in the MVRFA task, it can be improved by introducing more prior knowledge. It is difficult for the model to forecast the rare visual relationships and arrange them due to the long-tail annotations in real-world videos. By fully understanding the prior information, the model can link the unseen or rarely seen visual relationships with the current scene. In future work, we are going to introduce the prior information to assist in forecasting unbiased visual relationships.

CRediT authorship contribution statement

Wanping Ouyang: Investigation, Conceptualization, Software, Validation, Data curation, Writing - original draft, Visualization, Formal analysis. **Yaosi Hu:** Methodology, Conceptualization, Writing - review & editing, Supervision, Project administration. **Yangjun Ou:** Methodology, Conceptualization, Writing - review & editing, Supervision. **Zhenzhong Chen:** Methodology, Resources, Writing - review & editing, Supervision, Funding acquisition, Project administration.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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